

Design of Bandgap Current Reference

Name : Aryan Kumar

Roll no. : 230108010

Under guidance of
Dr. Mahima Arrawatia



Department of Electronics and Electrical Engineering

Indian Institute of Technology, Guwahati

Guwahati, 781039

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Bandgap Current Reference

Introduction

A bandgap current reference is a circuit that produces a very stable current that does not change much with temperature or supply voltage. It works by using two opposite effects in transistors: one voltage (V_{BE}) decreases as temperature increases, while another voltage difference (ΔV_{BE}) increases with temperature. By combining these two in the right way, their temperature effects cancel each other, giving a nearly constant voltage of about 1.2 V. This voltage is then passed through a resistor to create a steady current. Because of this, bandgap current references are widely used in electronic circuits where a reliable and stable current or voltage is needed, such as in voltage regulators and analog systems.

Design Procedure

Different Stages Involved in Circuit

- CTAT (Complement to Absolute Temperature)
- PTAT (Proportional to Absolute Temperature)
- Cascode Stage
- Startup Circuit

CTAT Circuit (Complement to Absolute Temperature)

The CTAT stage is the part of the bandgap circuit that produces a voltage which decreases as temperature increases. This voltage comes from the base-emitter junction of a transistor, called V_{BE} . As temperature rises, the internal behaviour of the transistor changes in such a way that V_{BE} gradually drops. This natural decrease is useful because it provides one side of the temperature compensation needed in the bandgap circuit. In simple terms, the CTAT stage gives a voltage that goes down with temperature, and this is later balanced by

another voltage that goes up with temperature to achieve an overall stable output.

Calculations :

Given:

$$V_{CTAT} = 0.7 \text{ V (Taken from Cadence)}$$

$$I_{ref} = 5 \text{ uA (current that we want to flow through CTAT)}$$

$$R_{CTAT} = V_{CTAT} / I_{ref}$$

$$R_{CTAT} = 0.7 / (5 \times 10^{-6})$$

$$R_{CTAT} = 140 \text{ k}\Omega$$

PTAT (Proportional to Absolute Temperature) Circuit

The PTAT circuit is the part of the bandgap reference that produces a voltage which increases as temperature increases. It is usually made using two transistors that are operated at different current densities (for example, by making one transistor larger than the other). Because of this difference, a small voltage difference called ΔV_{BE} is created between them. This voltage increases linearly with temperature. In simple terms, the PTAT circuit gives a voltage that goes up with temperature. This rising voltage is then scaled using resistors and combined with the CTAT voltage (which decreases with temperature) so that both effects cancel out, helping to create a stable and temperature-independent reference.

Calculations:

Given:

$$n = 8 \text{ (Emitter area ratio)}$$

$$V_T = 25 \text{ mV (Thermal voltage)}$$

$$I_{PTAT} = 5 \text{ uA (current that we want to flow through PTAT)}$$

$$\Delta V_{BE} = V_T \ln(n)$$

$$\Delta V_{BE} = 25 \times \ln(8)$$

$$\Delta V_{BE} \approx 25 \times 2.079$$

$$\Delta V_{BE} \approx 52 \text{ mV}$$

$$R = \Delta V_{BE} / I_{PTAT}$$

$$R = 52 \times 10^{-3} / (5 \times 10^{-6})$$

$$R \approx 10.4 \text{ k}\Omega$$

W/L Calculations for PTAT and CTAT circuits:

Given:

$$V_{ov} = 0.15 \text{ V } I_d = 10$$

$$\mu A_{unCox} = 150$$

$$\mu A/V^2 \mu pCox = 40$$

$$\mu A/V^2$$

$$L = 1 \text{ }\mu\text{m}$$

Formula used:

$$I_d = (1/2) * \mu C_{ox} * (W/L) * (V_{ov})^2$$

$$W/L = (2 * I_d) / (\mu C_{ox} * (V_{ov})^2)$$

For NMOS:

$$W/L = (2 * 10 \times 10^{-6}) / (150 \times 10^{-6} * (0.15)^2)$$

$$W/L = (20 \times 10^{-6}) / (150 \times 10^{-6} * 0.0225)$$

$$W/L \approx 5.93 \approx 6$$

Since $L = 1 \text{ }\mu\text{m}$:

$$W \approx 6 \text{ }\mu\text{m}$$

For PMOS:

$$W/L = (2 \times 10 \times 10^{-6}) / (40 \times 10^{-6} \times (0.15)^2)$$

$$W/L = (20 \times 10^{-6}) / (40 \times 10^{-6} \times 0.0225)$$

$$W/L \approx 22.2 \approx 22$$

Since $L = 1 \text{ } \mu\text{m}$:

$$W \approx 22 \text{ } \mu\text{m}$$

Cascode Stage

The right side of the circuit is a PMOS cascode current mirror. The upper PMOS transistor acts as the main current mirror device, which copies the reference current to the output branch. The lower PMOS transistor acts as the cascode device and is biased with a fixed voltage. This cascode transistor keeps the voltage across the upper mirror transistor almost constant, which helps in maintaining a steady current. By doing this, the circuit reduces the effect of output voltage changes and increases output resistance. In simple words, this stage ensures that the mirrored current remains accurate and stable.

Startup Circuit

A **3-transistor startup circuit** is used in bandgap references to make sure the circuit does not get stuck in a zero-current (OFF) state when power is first applied. Without this, the main circuit might never start because all transistors remain OFF and no voltage builds up.

The circuit uses three transistors working together. The first one is a **weak PMOS pull-up** that slowly raises a control node (often called START) towards the supply voltage. It is made weak so it only provides a small current and does not affect normal operation later.

The second transistor is an **NMOS** connected to a node inside the main circuit. It checks whether the core circuit has started working or not. When the circuit is OFF, this transistor is OFF. When the circuit turns ON and internal voltages rise, this transistor becomes ON.

The third transistor is another **NMOS** that actually starts the circuit. When the control node is high (due to the weak PMOS), this transistor turns ON and forces a key node in the main circuit to a value that turns ON the core transistors. This allows current to begin flowing and the circuit starts operating.

Schematic and DC operating Points

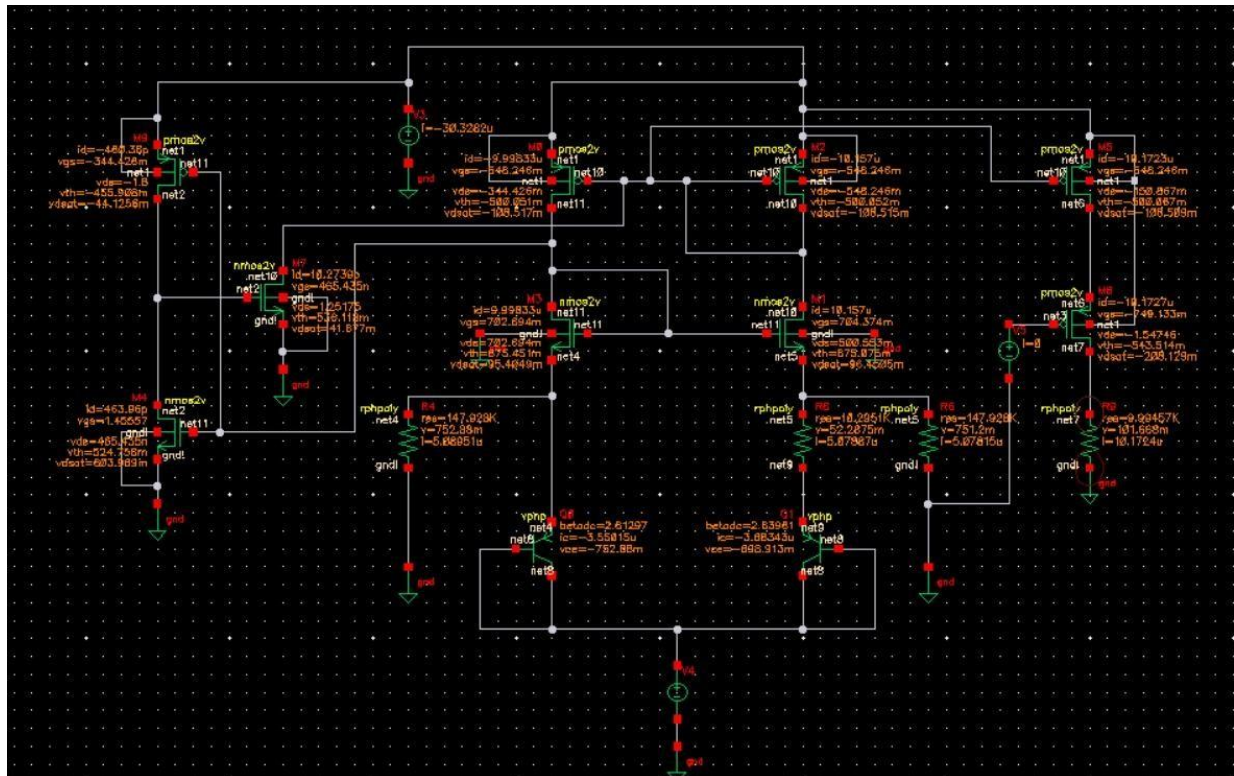


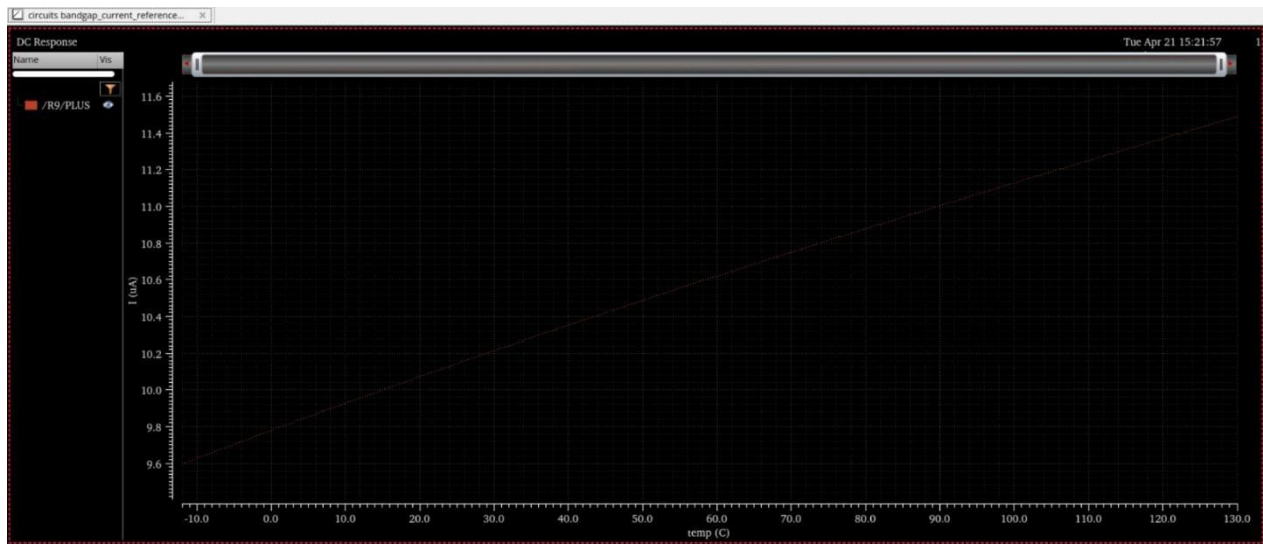
Fig. DC operating points

The DC operating point of this circuit shows the steady-state voltages and currents after the circuit has fully settled. From the values in your schematic, the main branch currents are around the desired μA range, which indicates that the reference current is properly established and mirrored across different branches.

The gate-to-source voltages of the transistors are higher than their threshold voltages, so all devices are turned ON. At the same time, the drain-to-source voltages are greater than the saturation voltages (V_{dsat}), which means the transistors are operating in the saturation region, as required for proper analog operation.

In the central part of the circuit, the BJT-based core is generating the expected base-emitter voltage (around 0.7 V) and a voltage difference between the two transistors, confirming that the CTAT and PTAT components are working correctly. On the right side, the PMOS cascode current mirror shows similar current levels in both cascoded devices, indicating proper current mirroring and improved output stability. On the left side, the startup circuit currents are very small, which means circuit is working properly. Overall, the DC operating point shows that the circuit is correctly biased, stable, and functioning properly.

Simulations



Output Current

The graph shows the variation of output current with temperature for your bandgap current reference. As temperature increases from -10°C to 125°C , the current increases gradually from about $9.6\text{ }\mu\text{A}$ to $11.4\text{ }\mu\text{A}$. Our graph has a positive slope, meaning the PTAT component is slightly dominating over the CTAT component, so the temperature effects are not perfectly cancelled. Ideally, the curve should have been flat, but in this case a small variation is present. However, the current still remains close to the target value of $10\text{ }\mu\text{A}$ across the entire temperature range, which shows that the circuit is functioning correctly.

Conclusion

After determination of all the parameters we have calculated power consumed in my circuit i.e.

Total current through the circuit is 30 μA and V_{dd} is 1.8 V so power consumed is $1.8 \times 30 = 54 \mu\text{W}$.

The bandgap current reference was successfully designed and implemented using CTAT and PTAT techniques along with a startup circuit and a PMOS cascode current mirror. The circuit achieves the desired reference current in the μA range, and DC operating point analysis confirms proper biasing and correct operation of all transistors. From the temperature analysis, the output current varies from 9.6 μA to 11.4 μA over the range of -10°C to 125°C , showing a moderate variation due to incomplete cancellation of temperature-dependent components. Despite this variation, the current remains close to the target value of 10 μA , indicating that the design is functioning correctly. Overall, the design has been successfully completed, and the results demonstrate a stable and reliable current reference with scope for further improvements.